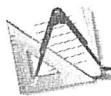


Few Ferri

第二章 函数的导数(Derivative)

2.1 导数的定义(The Definition of Derivative)



Multiple Choice Questions

C

1. Which of the following values is the average rate of change of $f(x) = \sqrt{x+2}$ over the interval $[-1, 2]$?

(A) -4

(B) $-\frac{1}{3}$

(C) $\frac{1}{3}$

(D) 3

$$\frac{2-1}{2+1}$$

C

2. Let $f(x) = 4 - 3x$. Which of the following is equal to $f'(0)$?

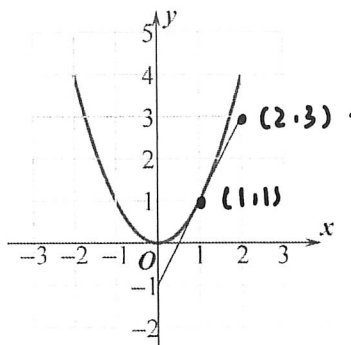
(A) 4

(B) -4

(C) -3

(D) 4

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1) = (1)$.



(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2

A

4. Let $f(x) = kx^3 - 2x$, ($k \neq 0$). The average rate of $f(x)$ over the interval $[-2, -1]$ is 3, find k .

(A) $\frac{5}{7}$

(B) $\frac{1}{7}$

(C) $-\frac{1}{7}$

(D) $-\frac{5}{7}$

$$\begin{aligned} & \{ + - 8k + 4 = -12 + 2 \\ & -1k = -5 \end{aligned}$$

C

5. **Calculator** Let $y = \tan e^x - x^2 + 1$. Find $\frac{dy}{dx} \big|_{x=0.5} =$

(A) -12.056

(B) 0

(C) 271.066

(D) 273.066

C

6. **Calculator** The graph of $y = \sqrt{x^2 + 1} - 3x$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

(A) 0.354

(B) -0.001

(C) -2.666

(D) 0

7. The population of insects at time t days is modeled by the function P with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?

- (A) At time $t = 10$, the population of insects is 230.
 (B) At time $t = 10$, the population of insects is increasing at a rate of 230 insects per day.
 (C) At time $t = 10$, the population of insects is decreasing at a rate of 230 insects per day.
 (D) At time $t = 10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function g at the point $(1, 4)$ passes through the point $(-2, -5)$, what is the slope of this line?

- (A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

Free Response Questions

9. Let $f(x) = x^2 - 2x$

(a) Find the average rate of change of f over $[1, 3]$.

$$\frac{\Delta f(x)}{\Delta x} = \frac{(3^2 - 2 \cdot 3) - (1^2 - 2 \cdot 1)}{3 - 1} = 2$$

(b) Find $\frac{f(2+h) - f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x = 2$.

$$(b) \frac{f(2+h) - f(2)}{h} = \frac{(2+h)^2 - 2(2+h) - 2^2 + 2 \cdot 2}{h} = \frac{4+h^2+4h-4-2h-4+4}{h} = \frac{h^2+2h}{h}$$

$$(c) \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} (h+2) = 2$$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2 \end{aligned}$$

$$f'(2) = 3 \times 2^2 = 12$$